



HR Analytics Report

An in-depth analysis of **17,414 employees** across **9 departments** — uncovering performance patterns, recognizing excellence, and identifying areas to improve workflow.

Dataset at a Glance

17,414

Total Employees

9

Departments

15

Features

Key feature groups: **Demographics** (Education, Gender, Age) • **Details** (Employee ID, Department, Region) • **Records** (Recruitment Channel, Trainings, Length of Service) • **Achievements** (Rating, KPIs, Awards, Training Score)

Data Preparation in Python

01

Load & Explore

Imported csv via Kaggle;
used `df.head()`, `df.info()`,
`df.isnull().sum()`

02

Clean Nulls & Duplicates

Education →
"Unknown";
`previous_year_rating` →
department mode;
dropped duplicates

03

Map & Engineer Features

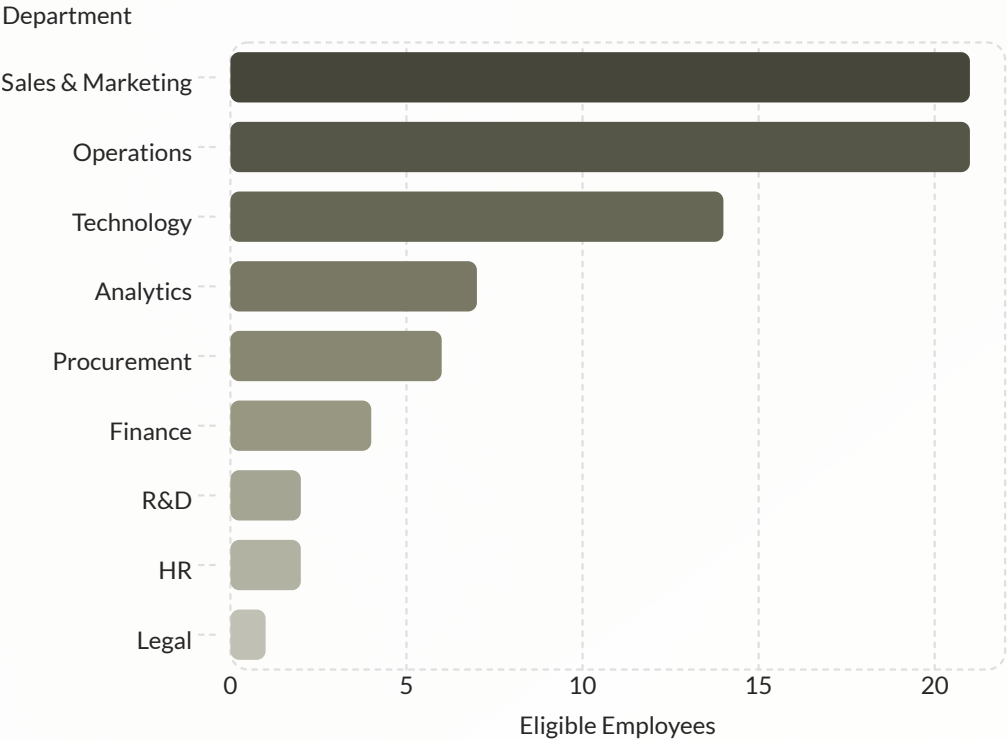
Gender: m→male,
f→female; created
`age_group` and
`work_experience`

04

Finalize

Renamed KPI feature name to lowercase characters; exported cleaned DataFrame to csv file.

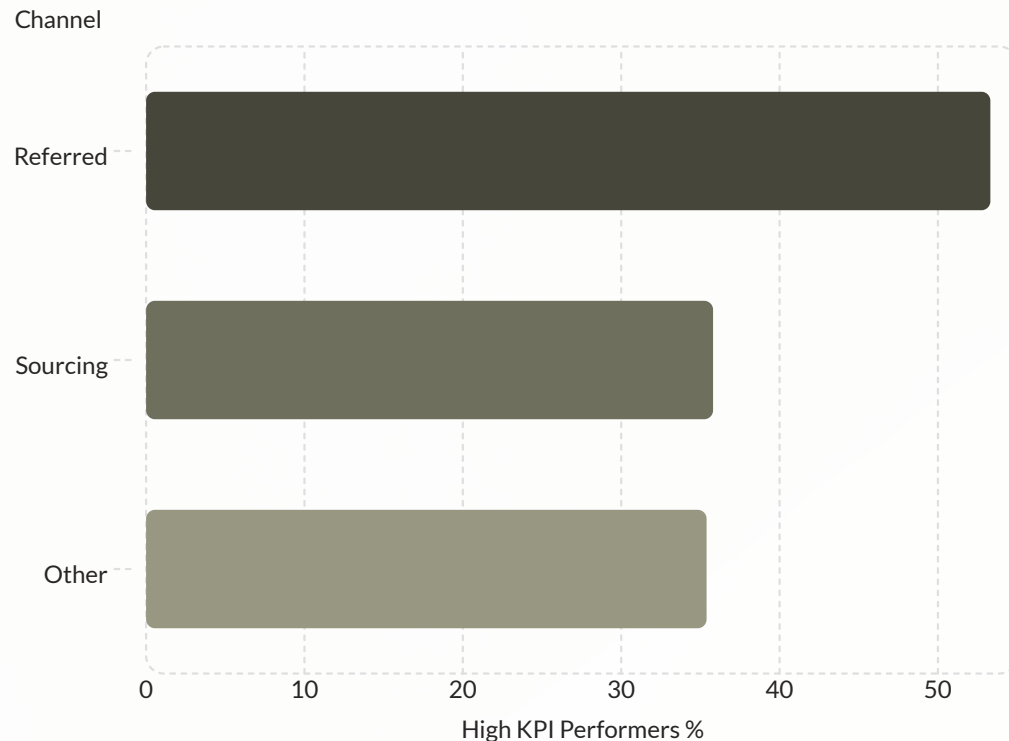
Top Performers by Department



Notable Averages

Technology leads in avg training score (**85.21**). Analytics employees have the longest avg tenure (**5.6 yrs**). R&D top performers average **85.50** training score despite only 2 eligible employees.

Recruitment Channel & Gender Insights



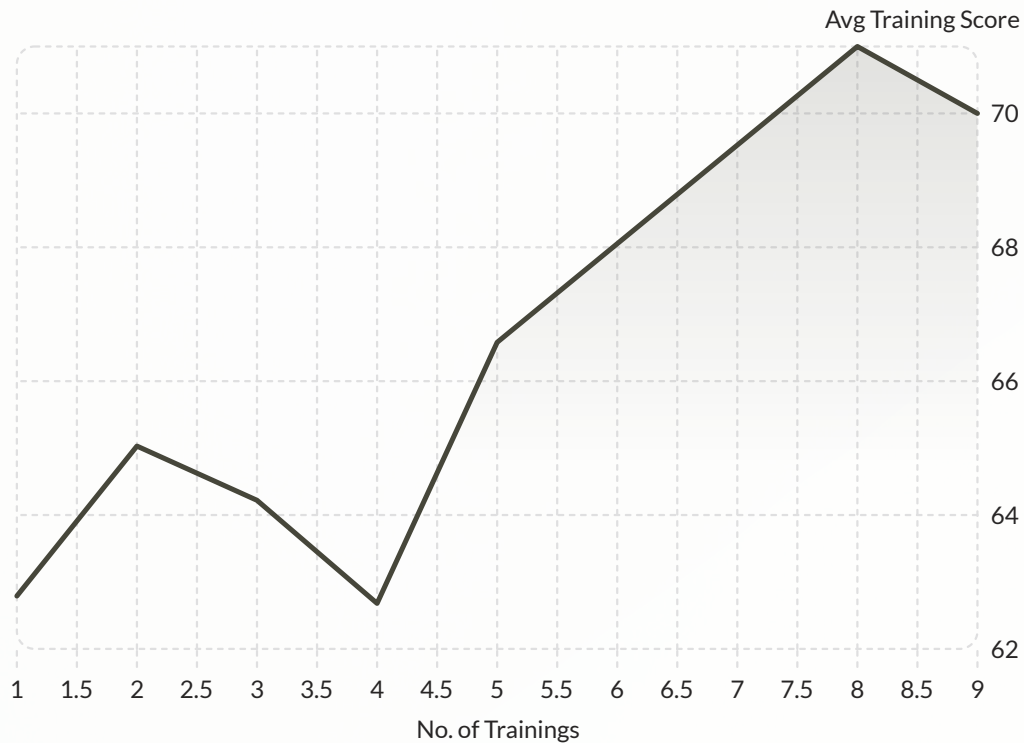
Recruitment

Referrals yield the highest-performing hires — **53.31%** meet KPIs vs. ~35% for other channels, despite fewer total hires (317 vs. 7,349 sourcing / 9,749 other).

Gender Distribution

Workforce is **71% male** and **29% female**.

Training Impact & Education Correlation



Training → Performance

More trainings generally raise avg training scores (peak at 8 trainings: **71.00**). However, KPI completion **declines** with more trainings — 1 training = 36.77% KPI met vs. 6 trainings = 0%.

Education & Tenure

Masters & above employees have the longest tenure (**7.4 yrs**). Below Secondary employees have shortest tenure (1.8 yrs) but surprisingly high training scores (65.90).

Struggling Employees & Performance Tiers



Where Struggles Concentrate

Sales & Marketing dominates struggling employees — especially region_2 (248 employees, avg training score 49.40).

Performance Tiers

Employees classified into tiers: **Tier 1: Top Performer** , **Tier 2: High Performer** , **Tier 3: Good Performer**, **Tier 4: Needs Improvement** — based on KPI, award won and rating.

Age, Experience & HR Metrics

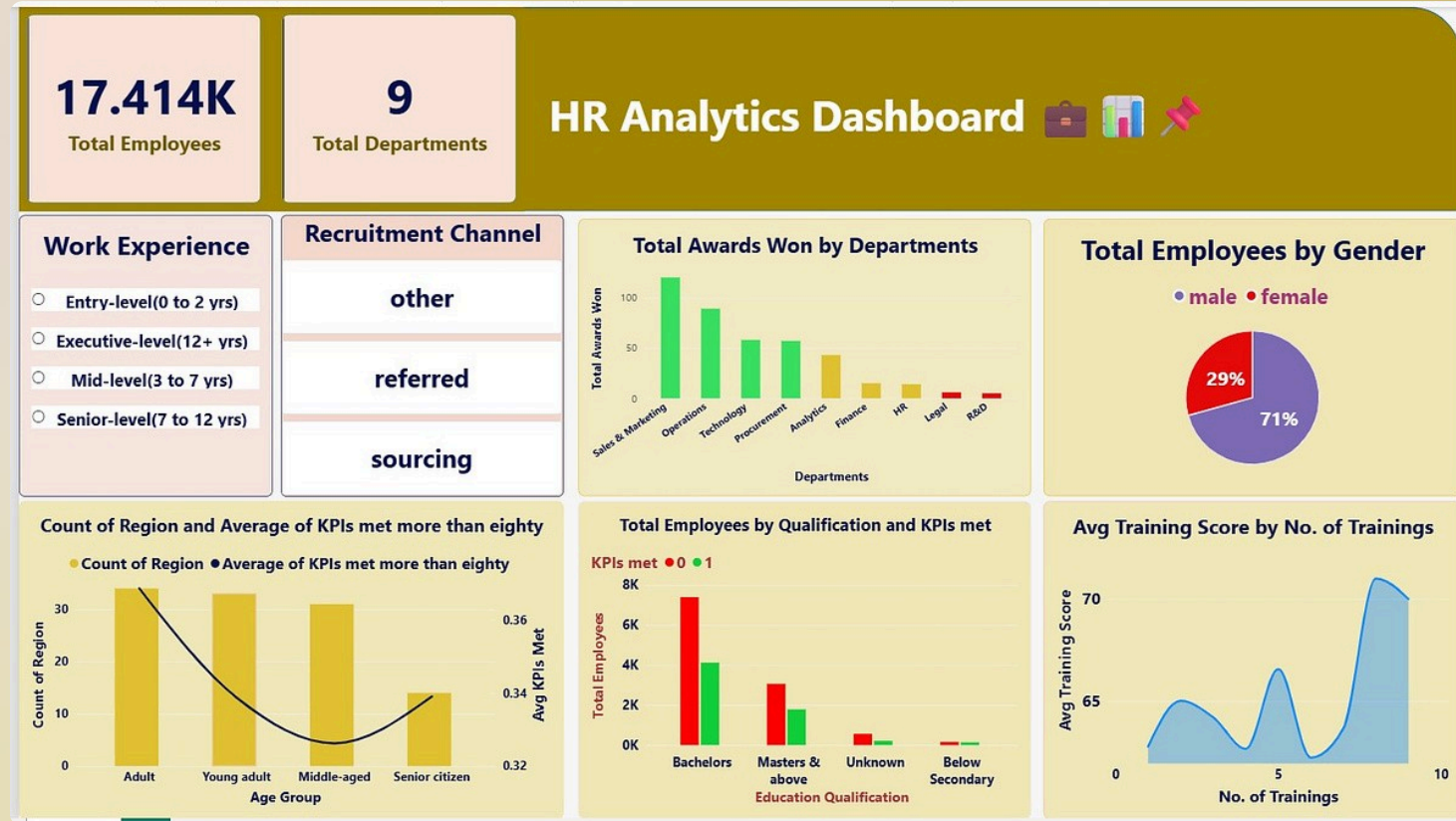
Age & Experience Profile

Adults dominate every department. Young adults have lowest ratings across departments except Legal Department.

Core HR Metrics

R&D leads in KPI met % (**44.88%**) and avg rating (3.7).
Analytics leads in avg training score (**84.57**). Sales & Marketing is largest dept (5,458 employees).

Power BI Dashboard



Key Insights & Recommendations

Key Insights

- More trainings → higher avg training scores
- Over half the workforce is male (71%)
- Referral channel yields maximum KPI-meeting employees
- KPI dips in <10 regions across young adults (entry/mid-level), middle aged(senior level) and adults(executive-level)

Recommendations

- Hire more capable **female** candidates for workforce balance
- **Entry & mid-level young adults** should focus on improving KPI goals
- Encourage **referral recruitment** — it consistently hires top performers
- **Executive-level adults** need to improve average KPI through greater goal engagement